Palm OS Developer Certification

Take a Training Course

The fastest method to learn the 'right way' to apply Palm OS technologies to the development of Palm applications and conduits is to take a course. Even if you have already written a few Palm OS applications, PalmSource offers courses to quickly boost your knowledge of topics such as communications, conduits, and advanced debugging. Developed by the experts at PalmSource, the training courses come in two formats: **online training** allows you to study at your own pace, and **instructor-led training**, taught at Palm OS Authorized Training Centers worldwide using authorized training materials and extensive hands-on labs. All of our courses teach and emphasize skills required for certification, covering the technologies tested in the exam.

If you are relatively new to Palm OS development, consider taking either the online course Quick Start to Palm OS ® Programming, or the four-day instructor-lead seminar Palm OS Programming Essentials. Both offerings provide extensive opportunities to apply your new skills with lab exercises and solution projects.

More experienced developers can extend their knowledge in several ways.

- The online seminar Palm OS Programming: Expert Case Studies examines common (and some not-so-common) programming challenges and techniques.
- Palm OS Communications Essentials focuses on extending applications via serial, low-level infrared, or network (sockets) communication APIs.
- Creating Palm OS Conduits (C/C++ on Windows) is a three-day course and is the fastest way to gain the knowledge you need to both create full-featured conduits and breeze through the conduit-related exam questions.
- Finally, consider Advanced Development and Debugging for Palm OS to improve your knowledge of such topics as code segmentation, shared libraries, standard IO applications, and Motorola MC68000 assembly debugging with the PalmDebugger.

Read the Documentation

Start with the Palm OS Programmer's Companion. PalmSource's official guide to application programming for Palm OS, this document contains conceptual and "how-to" information that complements the API Reference. Designed for random access, you can read the chapters in any order (the first few chapters are designed for programmers who are new to the Palm OS). Use the API Reference to look up function arguments and return values, but read the Companion to learn what functions to use and when. Volume II of the Companion focuses on communications. There are also Companion and Reference documents for conduit development on Windows and Mac OS.

Then, turn to the Palm OS Programming Recipes, technical articles with step-by-step explanations describing how to perform specific tasks related to Palm OS programming. The recipes differ from traditional sample code because they provide step-by-step explanations about common programming tasks. Recipes may contain text, code segments, sample code, or even downloadable project files.

Finally, there are numerous third-party books available for both novice and experienced Palm OS developers. Two books that provide broad coverage of Palm OS programming topics include:

- Palm OS Programming: The Developer's Guide, 2nd Edition by Neil Rhodes and Julie McKeehan. This book shows intermediate to experienced C and C++ programmers how to build a Palm application from the ground up. The book follows up the success of the first edition with expanded coverage of the Palm OS, including communications (serial and TCP/IP) and conduit development.
- Palm OS Programming Bible, 2nd Edition by Lonnon R. Foster is a complete reference on Palm programming written for the Palm OS up to version 5.0. It covers GUI application programming, web clipping applications, and conduits. It is written in a modular fashion so that you can easily read just the section you need for help.

Study the Sample Code

When learning how to apply the rich selection of Palm OS technologies, there is little substitute for examining working application source code. The source for the major PIM applications (such as Address and Date Book) as well as several games and utilities is included in the Palm OS software development kit (SDK). Some of these projects are relatively complex and difficult to follow by just scanning the source code, but two are particularly worth your consideration when looking for help.

- The Memo Pad application is the perfect blend of simplicity and completeness: it contains examples of many major Palm OS technologies (database, tables, categories, beaming and sending via the Exchange Manager, global Find, and fonts), but it also has a simple database format and user interface.
- The To Do application has a slightly more complex data format but also provides a wealth of code examples.

There are at least a couple of ways to profit from the samples provided with the SDK. Start by running the application on the Palm OS Emulator or Simulator and scan the source to examine how different features are implemented, getting a feel for the overall architecture of the project. Later, as you are reading the Palm OS Programmer's Companion (you *are* reading this, right?) or scanning the Palm OS Progammer's API Reference, use CodeWarrior's Find in Folders feature (or a utility such as grep or SpeedSearch X) to look for examples of the proper use of API functions.

Learn from Other Developers

You are not alone—the community of Palm OS developers is large and continues to grow at a rapid pace. This may seem like a tremendous challenge at first; after all, one of the top reasons for achieving certification is to allow your skills and creativity to stand out from the rest of the pack, and it is a very large pack. However, it is also a congenial pack, consisting of members who have backgrounds, skills, and questions not so different from yours. Whether you have a technical question about a particular part of the API or about some broader design topic, the developer forums are a great source of information. The forums are available as mailing lists (read and reply via email) or as newsgroups (read and reply via news reader). To prepare for the certification exam, you'll want to join and browse the Communications Developer Forum, the Conduit Forum, the Palm Developer Forum, the Tools Forum, and the Emulator Forum. All of these forums are archived and can be searched through the eScribe or eGroups web sites.